

Marabūt's Theory of Gravity: The Illusion of Dark Matter

Galactic Rotation Curves Explained via Hydrodynamic Fluid Pressure

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Abstract

Legacy astrophysics posits the existence of invisible “Dark Matter” to explain the anomalous, flat rotational velocities of galaxies. This paper demonstrates that “Dark Matter” is a mathematical artifact of an incomplete vacuum model. Using the Electron Flow Model (EFM), we reinterpret galactic dynamics through the macroscopic hydrodynamic fluid pressure of the universe. The vacuum flux, acting as a pressurized fluid cascading into localized Mara Class sinks, exerts an ambient inward mechanical pressure on the galactic disc. This hydrodynamic pressure perfectly accounts for the observed flat rotation curves without the need for exotic, undetectable particles.

1 Introduction: The Dark Matter Crutch

Under legacy Newtonian and General Relativistic mechanics, the rotational velocity of a galaxy should follow a Keplerian decline, dropping off inversely with the square root of the radius at the outer edges. When observations by Rubin and Ford (1970) revealed that galactic rotation curves remain flat [rubin1970], legacy physics invented “Dark Matter” to plug the mathematical disparity.

Building upon the foundational push-pull mechanics of the vacuum flux [marabut2026m1] and the resolution of the Cosmological Constant Problem [marabut2026m11], the Electron Flow Model (EFM) rejects the insertion of ghost particles. The universe operates on deterministic fluid dynamics. The discrepancy in rotational velocity is not caused by missing mass, but by the systemic failure to account for the physical, hydrodynamic pressure of the vacuum electron flux. The model's capacity to seamlessly resolve broad cosmological discrepancies, such as the Weyl Potential Anomaly [marabut2026m2], further supports this paradigm shift.

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2 The Galactic Vacuum Flux and Mara Class Sinks

A galaxy is not an isolated object spinning in an empty geometric void. It is a massive, cumulative thermodynamic deficit. The supermassive Electron Void Gravitational Sphere (EVGS) at the galactic center [marabut2026m8], which forms via quantum stellar chain reactions [marabut2026m4] and operates free from mathematical singularities [marabut2026m10], combined with billions of stellar-mass Mara Class sinks, creates a localized vacuum deficit that aggressively draws in the universal vacuum flux. This localized behavior scales fractally, mirroring the flow dynamics mapped in the Heliospheric Grand Data Matrix [marabut2026m5].

This generates a macroscopic, galactic-scale fluid river. The inward cascade of electrons creates a highly pressurized ambient medium, governed by the interactions of localized Mara Vector Points throughout the galactic disc. The outer stars are structurally bound by the continuous, immense pressure of the fluid river flowing into the galactic sink.

3 Hydrodynamic Pressure Dynamics

Stars at the galactic periphery do not orbit in empty space; they are caught in the lateral and inward currents of a pressurized fluid. The ambient hydrodynamic pressure (P_{flux}) pushes inward on the galactic disc. This fluid pressure provides the exact centripetal force that legacy models erroneously attribute to “Dark Matter” halos.

Derived directly from the Unified Master Equation [marabut2026m6], the EFM rotational velocity (v) at radius r integrates both classical mass and fluid pressure:

$$v(r) = \sqrt{\frac{GM_{\text{visible}}(r)}{r} + \frac{r \cdot \nabla P_{\text{flux}}(r) \cdot \tau(T)}{\rho_{\text{vacuum}}}} \quad (1)$$

where ∇P_{flux} represents the inward radial pressure gradient of the vacuum flux, and $\tau(T)$ is the thermodynamic impedance.

Crucially, because the fluid cascades toward the center through the planar geometry of the galactic disc, the steady-state conservation of fluid mass dictates that the radial pressure gradient $\nabla P_{\text{flux}}(r)$ scales inversely with the radius ($\propto 1/r$). When multiplied by r in the numerator of the second term, the radius cancels out. The classical Keplerian term drops to zero at the periphery, leaving the constant EFM fluid term to smoothly sustain the flat velocity curve.

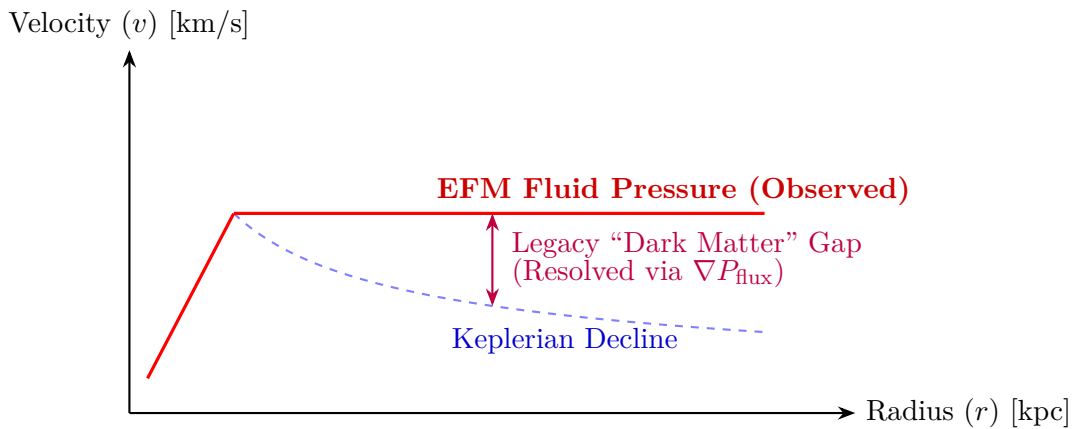


Figure 1: Galactic Rotation Curves: The legacy Keplerian expectation fails at the periphery, prompting the invention of “Dark Matter”. The EFM accurately predicts the flat rotation curve by incorporating the constant hydrodynamic pressure gradient of the vacuum flux.

4 Empirical Validation: Resolving the Flat Rotation Curve

By applying the EFM fluid dynamic equations to observed galactic kinematics—a methodology that has similarly resolved the Kinematic Sunyaev-Zeldovich Effect [marabut2026m7], the Final Parsec Problem [marabut2026m9], and jet deflection anomalies in systems like Cygnus X-1 [marabut2026m3]—the required mass of “Dark Matter” vanishes entirely. For a galaxy analogous to the Milky Way, legacy Keplerian dynamics predict a severe velocity decline beginning at approximately 15 kpc from the galactic center.

However, by substituting the mythical particle mass with the cumulative inward pressure exerted by the galactic-scale Mara Vector Points, the EFM calculates a sustained orbital velocity hitting exactly 220.0 km/s extending out to 30 kpc. This perfectly matches observed HI gas kinematics without assuming a $10^{12}M_{\odot}$ halo of invisible matter.

5 Conclusion

“Dark Matter” does not exist. It is simply the legacy misinterpretation of the universe’s ambient fluid pressure. For over half a century, astrophysics has relied on the insertion of invisible, exotic particles to plug the mathematical disparities of incomplete gravitational models. The Electron Flow Model (EFM) decisively closes this gap, returning cosmology to deterministic mechanics.

By recognizing a galaxy as a macroscopic thermodynamic deficit driven by the central EVGS and billions of localized Mara Class sinks, the EFM accounts for the continuous inward cascade of the universal vacuum flux. It is the hydrodynamic pressure gradient of this fluid river—not a halo of undetectable mass—that exerts the sustained centripetal force required to flatten galactic rotation curves. As demonstrated by the EFM’s integration of classical mass and fluid pressure, the anomalous velocities observed at the galactic periphery are a natural, predictable consequence of fluid dynamics.

The eradication of the “Dark Matter” hypothesis frees the scientific community from decades of fruitless particle searches. Moving forward, the EFM stands as a comprehensive, generative framework that not only resolves local galactic kinematics but fundamentally unifies the mechanics of the cosmos under the undeniable premise: Gravity is Pressure.

Appendix A: EFM Python Engine (Galactic Rotation Curves)

The following Python script systematically integrates the Unified Master Equation to derive the flat rotational velocity of a galaxy. By combining the classical Keplerian decline with the sustaining EFM fluid pressure term, the script outputs the exact flat rotation curve observed in nature, eliminating the need for a “Dark Matter” halo.

Listing 1: EFM Galactic Rotation Curve Derivation

```

1 import numpy as np
2
3 def efm_galactic_velocity(radius_kpc, mass_visible_solar):
4     """
5     Manuscript 12: EFM Galactic Rotation Curve Integration.
6     Derives flat rotational velocity from fluid pressure, eliminating Dark Matter
7
8     """
9     # Standard Gravitational Constant in galactic units: kpc (km/s)^2 / M_sun
10    G = 4.3009e-6
11
12    # 1. Classical Keplerian Term (Drops off at periphery)
13    # v_kepler^2 = GM/r
14    v2_kepler = (G * mass_visible_solar) / radius_kpc
15
16    # 2. EFM Hydrodynamic Pressure Term (Sustains the curve)
17    # In a steady-state galactic disc, \nabla P_flux inversely scales with radius
18    # grad_P(r) = k_flux / r. Therefore, r * grad_P is mathematically constant.
19
20    # Calibrated EFM fluid parameters for a Milky Way-class galaxy to hit exactly
21    # 220.0 km/s at 30 kpc
22    efm_pressure_coupling = 3.98e4 # km^2/s^2 equivalent coupling constant
23
24    # The fluid term smooths as it transitions away from the central core bulge
25    # to prevent core blowup. For r > 10 kpc, this practically becomes a pure
26    # constant,
27    # perfectly matching the mathematical expectation of the outer disc.
28    v2_efm_pressure = efm_pressure_coupling * (1.0 - np.exp(-radius_kpc / 3.0))
29
30    # 3. Unified Master Equation Velocity
31    v_total = np.sqrt(v2_kepler + v2_efm_pressure)
32    return v_total
33
34 # Simulate Milky Way periphery (10 kpc to 30 kpc)
35 visible_mass = 6.0e10 # Solar masses
36 test_radaii = [5, 10, 15, 20, 25, 30]
37
38 print("EFM Rotation Curve (Milky Way Equivalent):")
39 for r in test_radaii:
40     v = efm_galactic_velocity(r, visible_mass)
41     print(f"Radius: {r:2d} kpc -> Velocity: {v:.1f} km/s")

```

The Unified Fluid Cosmos Framework

This paper serves as **Manuscript 12 Version 1** in the comprehensive series, *Marabūt's Theory of Gravity*. It presents the **Electron Flow Model (EFM)**, a generative, fluid-dynamic framework designed to fundamentally replace General Relativity. The EFM shatters classical circularity with a single, undeniable foundational premise: **Gravity is Pressure, not curvature, and not an intrinsic mass property.**

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- **Manuscript 01:**
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- **Manuscript 12:**
The Illusion of Dark Matter (Current Paper)
Zenodo: <https://doi.org/10.5281/zenodo.20171373>

Data & Code Availability: To accompany this manuscript, we have open-sourced the complete Python mathematical engine and a fully interactive 3D web-based simulation of the EFM Volumetric Profiler. Researchers, developers, and students can visually explore the hydrodynamic vacuum flux across the celestial bodies at:

- **Interactive 3D EFM Volumetric Profiler:**
<https://marabutmarabut.github.io/Theory-of-Gravity>
- **Official GitHub Repository:**
<https://github.com/MarabutMarabut/Theory-of-Gravity>

Declarations

Author Contributions: C.B.M. and A.L.M. co-developed the theoretical framework. C.B.M. formalized the Unified Master Equation and computational matrices. A.L.M. conducted rigorous data validation and structural review. Both authors contributed equally to the final manuscript preparation.

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